

ISHAK SEDDIKI

Chemical Engineer

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PROFESSIONAL SUMMARY

Chemical R&D and Process Engineer with a Master's Degree in Refining Engineering and a Bachelor's Degree in Petroleum Engineering, combining process simulation and computational engineering. Experienced in refinery and chemical process modeling using Aspen HYSYS, DWSIM, and COMSOL Multiphysics, energy consumption, and production data. Skilled in machine learning, deep learning and engineering decision-making. Experienced in integrating simulation, laboratory experimentation, numerical modeling, and 3D engineering design to develop sustainable and high-performance solutions. Strong analytical, problem-solving, research, and technical skills.

PROFESSIONAL EXPERIENCE

Sonatrach Gas Treatment Division | Process Engineering Intern

MAR 2026 – APR 2026

Oued Noumer, Ghardaïa, Algeria

- Supported engineers and operators during gas-treatment activities, reviewed operating data and contributed to process monitoring and RVP optimization tasks.
- Gained field exposure to separation systems, heat exchangers, compressors, pumps, process control, utilities and safe industrial operations.
- Collaborated with multidisciplinary teams, followed site procedures and strengthened practical skills in technical communication, observation and daily operational reporting.

Sonatrach Transport Operations Intern (Laboratory of Analyses)

Dec2024 – Jan2025

In Amenas, Illizi, Algeria

- Assisted in laboratory analysis and quality control of petroleum and hydrocarbon products.
- Monitored and recorded operational and transportation data to support process efficiency and safety.
- Participated in sampling, testing, and analysis of oil and gas products in accordance with industry standards.
- Supported the evaluation of transport operations and logistics performance.
- Contributed to technical reporting, data interpretation, and documentation of laboratory results.

PERSONAL, PROCESS & R&D PROJECTS

Catalytic Methane–CO₂ Conversion to Syngas and Graphene/Carbon:

- Developed a COMSOL reactor model for dry reforming and carbon formation, coupling transport, reaction kinetics, thermal behavior and catalyst dependent performance. - Used the model as an automated evaluator to compare catalyst compositions and operating conditions for higher syngas and solid-carbon/graphene production.

CO₂ Battery & Packed-Bed Thermal Energy Storage

- Designed and simulated a 10 MW / 60 MWh CO₂ Battery in Aspen Plus, covering compression, liquefaction, pumping, heat recovery and turbine expansion.

-Modelled packed-bed TES in COMSOL and evaluated CO₂ hydraulics, heat transfer and transient behavior using PIPESIM and OLGA.

-Developed PFD/P&ID and plant layout in AutoCAD Plant 3D, including piping, valves, instrumentation and control elements.

Hydrocracking Process Modelling and AI Optimization:

- Built an Aspen HYSYS refinery workflow and evaluated reactor temperature, pressure, hydrogen-to-oil ratio, LHSV and recycle effects on conversion, yield and product quality. - Applied active learning and Bayesian/multi-objective optimization concepts to identify high-performance operating regions with fewer simulation evaluations.

Solarinc and Smart Solar Digital Twins

- Developed digital-twin concepts for solar operations and a smart solar tracker, integrating real-time data, IoT, computer vision and machine-learning based performance forecasting. - Used meteorological data and What-If scenarios to support energy-yield prediction, operational monitoring and predictive decision making.

AI-Driven Zeolite Synthesis Prediction

-Developed an end-to-end materials-informatics and machine-learning workflow to predict zeolite framework topology from precursor-gel composition, hydrothermal synthesis conditions and OSDA molecular structure. Engineered molecular features from SMILES descriptors, integrated heterogeneous experimental synthesis records, and applied data cleaning, iterative imputation and feature selection to generate a consistent ML-ready dataset. Built the framework to support zeolite topology classification, synthesis-condition analysis and data-driven screening of candidate synthesis routes.

Completed Refinery Process Modeling & Simulation – Aspen HYSYS

Developed an integrated Aspen HYSYS refinery model covering CDU, VDU, hydrocracking, FCC, visbreaking, NHT, reforming, isomerization and alkylation for an industrial-scale crude feed.

Evaluated conversion, product quality, hydrogen integration and refinery-wide material flows, creating a basis for future optimization, economic/environmental assessment and digital-twin development.

EDUCATION

Master of Process Engineering – Petroleum Refining | Faculty of Hydrocarbons and Chemistry, University of Boumerdes | 2022–2024

Relevant study: gas treatment, process modelling, refining, industrial thermodynamics, equipment, oil-and-gas analysis and process safety.

Bachelor of Process Engineering | University of Boumerdes | 2019–2022

Relevant study: water treatment, heat transfer, thermodynamics, distillation, kinetics, organic chemistry and process simulation.

CORE COMPETENCE

Process & Multiphysics

Aspen HYSYS, Aspen Plus, COMSOL, PIPESIM, OLGA, AutoCAD Plant 3D, mass/energy balances, reactor and equipment modelling, PFD/P&ID. gas processing, refining, CO₂ utilization, syngas, carbon materials, thermal storage, renewable energy and decarbonization.

AI, Data & Optimization

Python, machine learning, data cleaning, feature engineering, DOE/RSM, active learning and Excel.

R&D & Product Delivery

Technology scouting, requirements definition, prototype development, validation, technical documentation and stakeholder demonstr

Languages

Arabic – Native | English – Intermediate | French – Intermediate

